MACHINE LEARNING

MODEL METRICS

Domestic Flight Price Prediction

Github: https://github.com/ParthRawat1302/Machine-learning/tree/main/Domestic-Flight-Regression

1. How to Read These Numbers

- **RMSE** (root mean squared error): the evaluation metric; lower is better. Large errors on expensive tickets dominate it
- **R^2**: share of the price variance explained; higher is better
- **MAE** (mean absolute error): the typical error in rupees; lower is better
- **Economy / Dom. Biz / For. Biz**: RMSE inside each segment (Economy, domestic-carrier Business, foreign-carrier Business). Foreign Business is 10% of rows but most of the error
- **CV**: 5-fold cross-validation (KFold, shuffle, seed 42) on all 15,589 training rows, the same folds for every model. Every row is scored by a model that never saw it
- Blends with fitted weights are scored by nested CV (weights fitted on four folds, applied to the fifth)

2. Summary

The final model, an equal average of the 10 best models with a twin pull, reaches a CV RMSE of **9,474** (R^2 0.9414, MAE 2,972). Best configuration of each family:

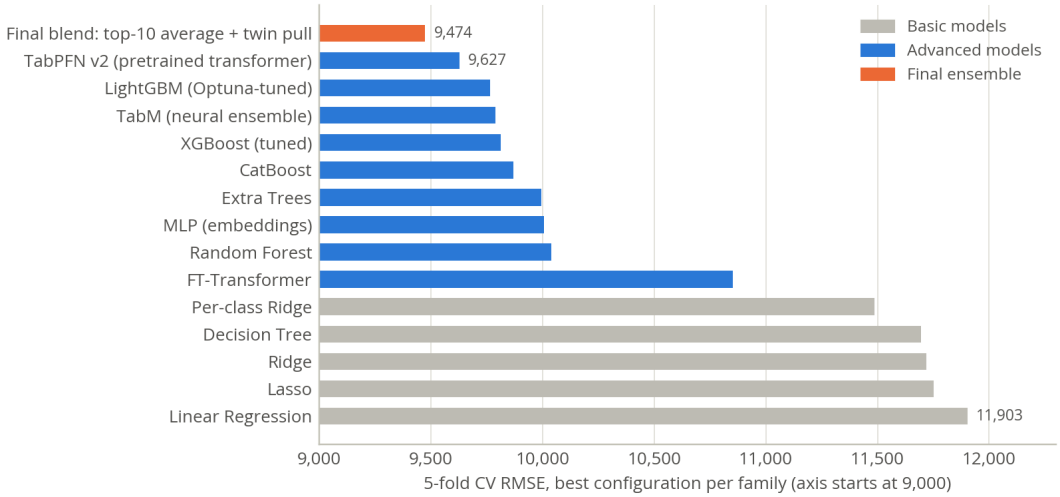


Figure 1. Best configuration of every model family, 5-fold CV RMSE.

3. Basic Models (5-fold CV)

Model	Features	Setting	CV RMSE	R^2	MAE
Linear Regression	one-hot	–	13,272	0.8849	6,151
Polynomial Regression	one-hot	degree 2	13,226	0.8857	6,128
Ridge	one-hot	α = 1	13,164	0.8868	6,068
Lasso	one-hot	α = 1	13,248	0.8854	6,123
Linear Regression	engineered	–	11,903	0.9075	5,300
Polynomial Regression	engineered	degree 2	11,956	0.9066	5,248
Ridge	engineered	α = 3	11,718	0.9103	5,253
Lasso	engineered	α = 3	11,750	0.9098	5,221
Decision Tree	one-hot	depth 12, leaf 5	11,694	0.9107	4,009
Decision Tree	engineered	depth 10, leaf 5	11,872	0.9079	4,230
Per-class Ridge	L5 ladder	α = 0.1, one model per cabin	11,486	0.9138	4,559

3.1 Segment RMSE of the basic models

Model	Features	Econ.	Dom. Biz	For. Biz
Linear Regression	one-hot	8,176	9,835	32,593
Polynomial Regression	one-hot	8,134	9,720	32,575
Ridge	one-hot	8,020	9,816	32,405
Lasso	one-hot	8,052	10,056	32,470
Linear Regression	engineered	4,176	10,252	31,372
Polynomial Regression	engineered	4,247	10,318	31,451
Ridge	engineered	3,893	10,281	30,860
Lasso	engineered	3,901	10,313	30,943
Decision Tree	one-hot	2,170	8,795	33,093
Decision Tree	engineered	2,331	8,936	33,541
Per-class Ridge	L5 ladder	2,415	9,347	31,776

3.2 Tuning sweeps

Ridge and Lasso CV RMSE for each penalty α (columns):

Model	Features	0.1	0.3	1	3	10	30	100	300	1000
Lasso	engineered	–	–	11,781	11,750	11,920	12,345	13,467	15,915	19,676
Lasso	one-hot	–	–	13,248	13,337	13,564	14,001	15,247	18,149	21,128
Ridge	engineered	11,854	11,806	11,744	11,718	11,865	12,307	13,305	–	–
Ridge	one-hot	13,194	13,176	13,164	13,226	13,570	14,422	16,062	–	–

Decision tree CV RMSE for each maximum depth (best minimum leaf size of 1, 5, 10, 20 or 40):

Features	4	6	8	10	12	15	20	none
one-hot	14,504	12,717	12,005	11,707	11,694	11,711	11,799	11,707
engineered	14,667	12,972	12,209	11,872	12,004	12,133	12,178	12,243

4. Advanced Models (5-fold CV)

All 22 saved models, sorted by CV RMSE. Each uses the full feature set (142 columns, target statistics nested inside the folds); test predictions come from a refit on all training rows.

Model	Configuration	CV RMSE	R^2	MAE	Econ.	Dom. Biz	For. Biz
TabPFN v2	**per cabin, 8 members, seed 1**	**9,627**	**0.9395**	**2,976**	**1,857**	**6,751**	**27,601**
TabPFN v2	per cabin, + composite feats	9,644	0.9393	2,985	1,862	6,858	27,574
TabPFN v2	per cabin, 4 members, seed 0	9,644	0.9392	2,997	1,866	6,738	27,666
LightGBM	Optuna trial 9, 3 seeds	9,767	0.9377	3,168	1,715	6,839	28,071
LightGBM	Optuna trial 13, 3 seeds	9,774	0.9376	3,159	1,689	6,870	28,082
TabM (k = 32)	3 blocks × 256, 3 seeds	9,789	0.9374	3,376	1,858	6,982	27,986
LightGBM	trial 13 + all categoricals	9,793	0.9374	3,161	1,711	6,849	28,157
TabM (k = 32)	4 blocks × 512, 2 seeds	9,795	0.9373	3,580	2,066	7,006	27,905
LightGBM	Optuna trial 12, 3 seeds	9,812	0.9371	3,204	1,736	6,886	28,187
XGBoost	Optuna: depth 4, Tweedie	9,813	0.9371	3,244	1,769	6,844	28,208
CatBoost	depth 8, all categoricals	9,869	0.9364	3,315	1,786	6,913	28,345
CatBoost	depth 6, random strength 2	9,874	0.9363	3,360	1,839	6,917	28,337
TabPFN v2	per segment, 8 members	9,877	0.9363	3,071	1,850	7,311	28,041
CatBoost	depth 6, all categoricals	9,954	0.9353	3,378	1,826	6,877	28,652
Extra Trees	500 trees, min leaf 3	9,994	0.9348	3,211	1,748	6,969	28,749
XGBoost	hand-set: depth 10	9,997	0.9347	3,225	1,762	6,888	28,814
MLP + embeddings	3 × 256, 3 seeds	10,008	0.9346	3,691	2,040	7,233	28,481
Random Forest	500 trees, min leaf 3	10,040	0.9342	3,302	1,838	7,023	28,834
LightGBM	trial 13, Tweedie loss	10,402	0.9293	3,640	2,039	7,670	29,519
FT-Transformer	d = 96, 3 layers	10,854	0.9231	4,318	2,543	8,388	30,296
FT-Transformer	d = 64, 43 curated features	11,131	0.9191	4,904	3,254	8,350	30,923
Per-class Ridge	L5 features, α = 0.1	11,486	0.9138	4,559	2,415	9,347	31,776

5. Ensembles

Every blend tried, with and without the twin pull (prediction moved toward the price of exact twins in the training data; weight chosen by nested CV).

Blend	CV RMSE	R^2	MAE	Econ.	Dom. Biz	For. Biz
Average of the 10 best + twin pull 0.8	**9,474**	**0.9414**	**2,972**	**1,674**	**6,532**	**27,302**
Equal average of strong families + twin pull 0.8	9,487	0.9412	3,010	1,678	6,557	27,328
Average of the 5 best + twin pull 0.8	9,489	0.9412	2,918	1,687	6,548	27,339
Convex weights over family averages + twin pull 0.8	9,510	0.9409	2,996	1,713	6,568	27,384
Average of the best model per family + twin pull 0.8	9,520	0.9408	3,065	1,730	6,620	27,375
Hill-climbing selection + twin pull 0.8	9,529	0.9407	3,007	1,726	6,546	27,463
Convex weights, all models + twin pull 0.8	9,529	0.9407	3,008	1,726	6,546	27,464
Average of the 10 best	9,533	0.9406	3,038	1,681	6,686	27,390
Equal average of strong families	9,536	0.9406	3,079	1,685	6,665	27,414
NNLS weights + twin pull 0.8	9,537	0.9406	3,035	1,715	6,535	27,502
Average of the 5 best	9,543	0.9405	2,973	1,695	6,695	27,413
Convex weights per segment + twin pull 0.8	9,544	0.9405	2,989	1,660	6,579	27,515
Hill-climbing per segment + twin pull 0.8	9,545	0.9405	2,989	1,660	6,580	27,517
Convex weights over family averages	9,552	0.9404	3,057	1,721	6,680	27,444
NNLS weights per segment + twin pull 0.8	9,555	0.9404	3,048	1,655	6,562	27,566
Hill-climbing selection	9,576	0.9401	3,069	1,734	6,674	27,527
Convex weights, all models	9,576	0.9401	3,071	1,734	6,673	27,530
Average of the 3 best + twin pull 0.8	9,580	0.9400	2,932	1,845	6,646	27,521
NNLS weights	9,585	0.9400	3,098	1,723	6,661	27,572
Average of the best model per family	9,591	0.9399	3,149	1,739	6,784	27,495
Convex weights per segment	9,593	0.9399	3,048	1,667	6,716	27,578
Hill-climbing per segment	9,593	0.9399	3,048	1,667	6,717	27,580
NNLS weights per segment	9,603	0.9398	3,110	1,662	6,697	27,629
Average of the 3 best	9,616	0.9396	2,977	1,855	6,765	27,554

6. Team Notebook Experiments

Scores from the notebooks in *Domestic Flight Price Prediction Challenge/*. The holdout rows use one 80/20 split, which happens to contain fewer extreme fares, so they read lower than 5-fold CV.

Notebook model	Split	RMSE	MAE	R^2
Linear Regression, one-hot incl. flight	holdout	12,137	5,759	0.8955
Ridge α = 1, one-hot incl. flight	holdout	12,125	5,735	0.8957
Polynomial (degree 2) + one-hot	holdout	12,069	5,804	0.8967
Lasso α = 100, one-hot	holdout	14,251	7,202	0.8559
Linear Regression, engineered	holdout	11,025	5,111	0.9137
Ridge α = 1, engineered	holdout	10,986	5,081	0.9144
Extra Trees, 500 trees	holdout	10,193	3,128	0.9263
Extra Trees tuned (900 trees, min leaf 2)	holdout	10,166	3,125	0.9267
CatBoost (depth 8)	holdout	10,551	4,044	0.9210
XGBoost (depth 7, 900 trees)	holdout	9,977	3,629	0.9294
XGBoost tuned (depth 6, 1,100 trees)	holdout	9,872	3,688	0.9309
Blend 65% XGBoost + 35% Extra Trees	holdout	9,751	3,375	0.9325
XGBoost, flight-aware features	5-fold	10,386	3,851	0.9295
Extra Trees, flight-aware features	5-fold	11,202	3,274	0.9180
CatBoost, flight-aware features	5-fold	10,768	4,089	0.9243
Blend XGB / ET / CatBoost (OOF weights)	5-fold	10,159	3,576	0.9326

7. Hyperparameter Search (Optuna)

Each trial is a full 5-fold CV run. The five best trials of each search:

LightGBM (32 trials)

Trial	CV RMSE	objective	num leaves	min child samples	learning rate	columns per tree	extra trees
13	9,772	regression	196	18	0.032	0.472	True
9	9,779	regression	196	4	0.044	0.469	True
21	9,779	regression	105	34	0.031	0.451	True
12	9,787	regression	105	24	0.043	0.487	False
31	9,799	regression	74	50	0.041	0.527	True

XGBoost (20 trials)

Trial	CV RMSE	objective	max depth	min child weight	eta	columns per tree	subsample
17	9,784	tweedie	4	10.377	0.036	0.512	0.782
1	9,798	tweedie	5	8.369	0.043	0.619	0.893
14	9,829	tweedie	4	5.084	0.032	0.837	0.883
4	9,834	tweedie	5	6.086	0.038	0.991	0.886
18	9,847	tweedie	5	5.800	0.028	0.767	0.844

8. Feature Ablation (LightGBM, 5-fold CV)

Step	CV RMSE	Change
Native categoricals only	10,320	–
+ target statistics	10,373	+53
+ day effects	10,191	−182
+ weekday effects	10,104	−87
+ bigger trees (63 leaves)	9,958	−146
+ flight-day effects	9,909	−49
+ Optuna tuning (trial 13, 1 seed)	9,772	−138

9. Training-Set Size (LightGBM, 5-fold CV)

Share of rows used for training	CV RMSE	Change
48%	10,784	–
64%	10,262	−522
80% (a full training fold)	9,909	−353

10. Other Variants Tried During Development

Rejected or superseded configurations, scored on the same 5-fold CV but with the feature set available at the time, so compare each row with its own baseline column.

Variant	Features at the time	CV RMSE	Baseline
LightGBM, log(price) target	target statistics	10,412	10,329 (raw price)
LightGBM, residual on shrunk itinerary mean	target statistics	10,547	10,329
LightGBM, one model per cabin	target statistics	10,437	10,329
LightGBM, heavy regularisation	+ day effects	10,316	10,191
LightGBM, Poisson objective	+ weekday effects	10,106	10,104
LightGBM, Tweedie objective (1.5)	+ weekday effects	10,044	10,104
LightGBM, kernel-smoothed day effects	final	9,982	9,909
XGBoost, depth 6 (GPU)	target statistics	10,380	–
CatBoost, all categoricals	+ day effects	10,151	–
MLP, embeddings for all categoricals	+ day effects	10,568	10,351 (low-card only)
MLP, 512 wide, dropout 0.3, all categoricals	+ day effects	10,634	10,568 (256 wide)
TabM, piecewise-linear embeddings	+ weekday effects	10,583	9,951 (no embeddings)
TabM, embeddings, learning rate 5e-4	+ weekday effects	10,239	9,951
TabM, all categoricals one-hot	+ weekday effects	10,229	9,951
TabM, single seed (3 sizes)	+ weekday effects	9,951–9,975	9,789 (3 seeds, final features)